

$$U(\vec{0}) = U_L(\theta_e) \cdots U_e(\theta_e) \cdots U_1(\theta_e) = W U_e(\theta_e) V$$

$$\begin{aligned} C(\vec{0}) &= \langle \psi_0 | U^\dagger(\vec{0}) \hat{O} U(\vec{0}) | \psi_0 \rangle = \langle \psi_0 | V^\dagger U_e^\dagger(\theta_e) \underbrace{W^\dagger \hat{O} W}_A \underbrace{U_e(\theta_e) V}_{1\psi} | \psi_0 \rangle \\ &= \langle \psi | U_e^\dagger(\theta_e) A U_e(\theta_e) | \psi \rangle \\ &= \langle \psi | R^\dagger(\theta_e) A R(\theta_e) | \psi \rangle \end{aligned}$$

$$U_e(\theta_e) = R(\theta_e) = e^{-i \frac{\theta_e}{2} P} \leftarrow$$

$$= \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} P$$

$$\frac{\partial C}{\partial \theta_e} = \frac{i}{2} \langle \psi | R^\dagger(\theta_e) P A R(\theta_e) - R^\dagger(\theta_e) A P R(\theta_e) | \psi \rangle$$

$$\frac{\partial R(\theta_e)}{\partial \theta_e} = -i \frac{P}{2} R(\theta_e) \leftarrow$$

$$\frac{i}{2} \langle \psi | R^\dagger(\theta_e) [P, A] R(\theta_e) | \psi \rangle$$

$$\frac{1}{2} \langle \psi | R^\dagger(\theta_e) \left(R^\dagger\left(\frac{\pi}{2}\right) A R\left(\frac{\pi}{2}\right) - R\left(\frac{\pi}{2}\right) A R^\dagger\left(\frac{\pi}{2}\right) \right) | \psi \rangle \neq R^\dagger(\psi) A R$$

$$= \cos^2 \frac{\theta}{2} I + \sin^2 \frac{\theta}{2} P A P + \frac{i}{2} \sin \theta [P, A]$$

$$= \frac{1}{2} \langle \psi | R^\dagger(\theta_e + \frac{\pi}{2}) A R(\theta_e + \frac{\pi}{2}) | \psi \rangle$$

$$R(\psi) A R^\dagger(\psi) =$$

$$- \frac{1}{2} \langle \psi | R^\dagger(\theta_e - \frac{\pi}{2}) A R(\theta_e - \frac{\pi}{2}) | \psi \rangle$$

$$= \cos^2 \frac{\theta}{2} I + \sin^2 \frac{\theta}{2} P A P - \frac{i}{2} \sin \theta [P, A]$$

$$\frac{1}{2} [C(\vec{0}, \theta_e + \frac{\pi}{2}) - C(\vec{0}, \theta_e - \frac{\pi}{2})]$$

$$R^\dagger(\psi) A R - R(\psi) A R^\dagger(\psi)$$

$$= i \sin \theta [P, A] \quad \theta = \frac{\pi}{2}$$

$$(2 \times \text{shots} \times L \times M)$$

$$\rightarrow i[P, A]$$

$$H|E_n\rangle = E_n|E_n\rangle, \quad E_0 \leq E_1 \leq \cdots \leq E_{N-1}$$

$$|\psi\rangle = \sum_n c_n |E_n\rangle$$

$$\langle \psi | H | \psi \rangle = \left(\sum_i c_i^* \langle E_i | \right) \left(\sum_n E_n |E_n\rangle \langle E_n| \right) \left(\sum_j c_j |E_j\rangle \right)$$

$$= \sum_n c_n^* c_n E_n$$

$$\langle E_1 | E_2 \rangle = 0$$

$$= \sum_n P_n E_n = E_0 + \sum_n P_n \Delta E_n \geq E_0$$

$$E_n = E_0 + \underbrace{\Delta E_n}_{\geq 0}$$

$$H = -J \sum_{i=0}^{N-1} z_i z_{i+1} - h \sum_{i=0}^{N-1} x_i, \quad |\psi\rangle = \prod_{i=0}^{N-1} R_i(\theta) |0\rangle^{\otimes N}$$

$$C(\theta) = \frac{\langle \psi | H | \psi \rangle}{N}$$

$$= (\cos \frac{\theta}{2} |0\rangle + \sin \frac{\theta}{2} |1\rangle)^{\otimes N}$$

$$\equiv |0\rangle^{\otimes N}$$

$$= -\frac{J}{N} \sum_{i=0}^{N-1} \langle \psi | z_i z_{i+1} | \psi \rangle - \frac{h}{N} \sum_{i=0}^{N-1} \langle \psi | x_i | \psi \rangle$$

$$\langle 0 | X | 0 \rangle = (\cos \frac{\theta}{2} \langle 0 | + \sin \frac{\theta}{2} \langle 1 |) X (\cos \frac{\theta}{2} |0\rangle + \sin \frac{\theta}{2} |1\rangle)$$

$$X = |0\rangle\langle 1| + |1\rangle\langle 0|,$$

$$X|0\rangle = |1\rangle$$

$$\langle 0 | 1 \rangle = 0$$

$$\langle 0 | 0 \rangle = 1$$

$$X|1\rangle = |0\rangle$$

$$\langle 0 | X | 0 \rangle = (\cos \frac{\theta}{2} \langle 0 | + \sin \frac{\theta}{2} \langle 1 |) (\cos \frac{\theta}{2} |1\rangle + \sin \frac{\theta}{2} |0\rangle)$$

$$\downarrow = \cos \frac{\theta}{2} \sin \frac{\theta}{2} + \cos \frac{\theta}{2} \sin \frac{\theta}{2} = \sin \theta$$

$$\langle 00 | z_0 z_1 | 00 \rangle = \langle 0 | z_0 | 0 \rangle \langle 0 | z_1 | 0 \rangle \quad \checkmark$$

$$\langle 0 | z | 0 \rangle = (\cos \frac{\theta}{2} \langle 0 | + \sin \frac{\theta}{2} \langle 1 |) z (\cos \frac{\theta}{2} |0\rangle + \sin \frac{\theta}{2} |1\rangle)$$

$$z = |0\rangle\langle 0| - |1\rangle\langle 1|, \quad z|0\rangle = |0\rangle \quad z|1\rangle = -|1\rangle$$

$$\langle 0 | z | 0 \rangle = (\cos \frac{\theta}{2} \langle 0 | + \sin \frac{\theta}{2} \langle 1 |) (\cos \frac{\theta}{2} |0\rangle - \sin \frac{\theta}{2} |1\rangle)$$

$$\cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} = \cos \theta$$

$$\langle 00 | z_0 z_1 | 00 \rangle = \cos^2 \theta$$

$$C(\theta) = -\frac{J}{N} \sum_{i=0}^{N-1} \cos^2 \theta - \frac{h}{N} \sum_{i=0}^{N-1} \sin \theta$$

$$c(\theta) = -J \cos^2 \theta - h \sin \theta = \frac{\langle \psi | H | \psi \rangle}{N}$$

$$\frac{\partial c(\theta)}{\partial \theta} = 0 = 2J \cos \theta \sin \theta - h \cos \theta = 0$$

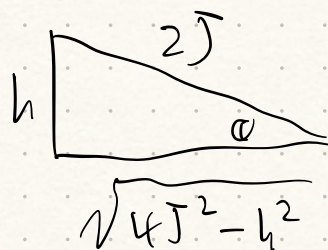
$$\textcircled{a} \quad \cos \theta (2J \sin \theta - h) = 0$$

$$\cos \theta = 0 \quad \text{or} \quad 2J \sin \theta - h = 0 \quad \textcircled{b}$$

$$\Rightarrow \sin \theta = \frac{h}{2J}$$

$$\frac{h}{2J} \leq 1$$

$$h \leq 2J$$



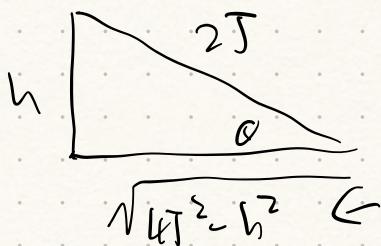
$$\textcircled{a} \quad c(\theta^*) = -h$$

$$\textcircled{b} \quad c(\theta^*) = -J \frac{(4J^2 - h^2)}{4J^2} - \frac{h^2}{2J}$$

$$-J - \frac{h^2}{4J}$$

$$\cos \theta = \frac{\sqrt{4J^2 - h^2}}{2J}$$

$$\frac{\langle \psi | H | \psi \rangle}{N} = \begin{cases} -J^2 - \frac{h^2}{4J} & \text{if } 0 \leq h < 2J \\ -h & \text{if } h \geq 2J \end{cases}$$



$$m = \frac{1}{N} \sum_{i=0}^{N-1} \langle \psi | z_i | \psi \rangle$$

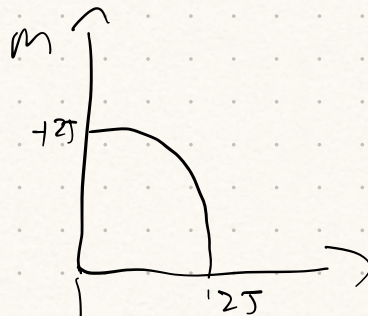
$$= \cos \theta$$

$$m = \frac{1}{2J} \sqrt{4J^2 - h^2}$$

$$\frac{\sqrt{4J^2 - h^2}}{2J} = m$$

$$m^2 = \frac{4J^2 - h^2}{4J^2} \Rightarrow$$

$$m^2 + h^2 = 4J^2$$



$$\hbar^2 - \hbar^2 = \hbar^2 m^2$$

$$\hbar^2 = \hbar^2 m^2 + \hbar^2$$

$$1 = m^2 + \frac{\hbar^2}{\hbar^2} \rightarrow g^2$$

$$1 = m^2 + g^2$$

